SF2943 Project — Part A

Daily electricity load for Swedish bidding zone SE_3

1 Data

We use daily mean electricity load (in MW) for the Swedish bidding zone **SE_3** (Stockholm and the Mälardalen region in eastern Sweden). The data come from *Open Power System Data* (OPSD), the file `time_series_60min_singleindex.csv`, column `SE_3_load_actual_entsoe_transparency`. The original source is the ENTSO-E Transparency Platform.[1] The file contains hourly observations from 2015-01-01 01:00 to 2020-09-30 22:00 UTC (50 398 hours, 77 NaN, longest gap 49 h). We aggregate to daily means and mark days with fewer than 20 valid hours as NaN. The 5 remaining NaN days are filled by linear interpolation. This gives $n = 2100$ daily observations X_t, which is well above the $N \geq 2000$ requirement.

2 Cleaning to stationarity

The raw series X_t has a clear winter/summer cycle, a weekday/weekend cycle, a small long-term trend, and an amplitude that grows with the level. We use the classical decomposition (B&D §1.5) on the log of the series:

$$\log X_t \;=\; m_t \;+\; s_t^{(\text{year})} \;+\; s_t^{(\text{week})} \;+\; Y_t, \tag{1}$$

fitting in the order *year* → *week* → *trend* by ordinary least squares (OLS). Each step is fit on the residual of the previous one. Taking logs (Box–Cox with $\lambda = 0$) stabilises the variance and turns multiplicative seasonality into additive, which is what makes (**??**) reasonable.

Yearly seasonality. A $k = 2$ harmonic (B&D §1.3) with period $d = 365$:

$$\hat{s}_t^{(\text{year})} \;=\; \sum_{j=1}^{2} \big[a_j \cos(2\pi j t/365) + b_j \sin(2\pi j t/365)\big],$$

with $(a_1, b_1, a_2, b_2) = (+0.2384, +0.0773, -0.0127, -0.0013)$. The first harmonic catches the main winter–summer swing and the second one handles the asymmetric shape of the peak.

Weekly seasonality. A $k = 3$ harmonic with period $d = 7$, fit on the year-deseasonalised series. The coefficients are $(a_1, \ldots, b_3) = (+0.0323, -0.0536, +0.0096, +0.0304, -0.0086, -0.0107)$. Since $d = 7$ is odd, $k = 3$ is the saturated Fourier basis for the weekly cycle, which is equivalent to using six day-of-week dummies.

Polynomial trend. We start with degree 1 on the deseasonalised series. If the residual mean of any third of the series is bigger than 10% of the residual standard deviation, we move up to degree 2. Here the linear residual thirds were $(-0.0061, +0.0113, -0.0052)$ with std 0.0645 (max ratio ≈ 0.18), so we picked the quadratic

$$\hat{m}_t \;=\; 9.1609 \;+\; (6.12 \times 10^{-5})\, t \;-\; (3.35 \times 10^{-8})\, t^2.$$

The cleaning residual $\hat{Y}_t = \log X_t - \hat{m}_t - \hat{s}_t^{(\text{year})} - \hat{s}_t^{(\text{week})}$ is shown in Figure **??**c. By eye the level and amplitude look constant.

Diagnostics. Ljung–Box clearly rejects iid noise ($Q(20) = 4134$, $p \approx 0$), so cleaning has removed the deterministic part but there is still autocorrelation left for the ARMA stage to handle. The sample ACF (Figure **??**a) decays slowly with $\hat{\rho}(1) = 0.83$, has a clear spike at lag 7 ($\hat{\rho}(7) = 0.30$), and a small leftover yearly signal at $\hat{\rho}(365) = 0.16$. The PACF (Figure **??**b) has its biggest spike at lag 1 with smaller activity at lags 2–3, which suggests an AR-dominated process. As a sanity check, $\nabla_1 \nabla_7 \log X_t$ (B&D §1.5.2) gives a similar-looking ACF.

3 ARMA fitting

Based on the ACF/PACF shape we tried five candidate models, fit by Gaussian MLE (B&D §5.2; `statsmodels.tsa.arima.model.ARIMA` maximises the likelihood numerically using the innovations algorithm). All five fits are causal and invertible, i.e. all roots of $\hat{\phi}(z)$ and $\hat{\theta}(z)$ are outside the unit circle.

[1] `https://data.open-power-system-data.org/time_series/`

We pick the order using AICC (B&D §5.5.2), AICC = AIC + $2k(k+1)/(n-k-1)$ with $k = p+q+2$ counting the constant and σ^2. AICC values (params = $p+q$): ARMA(2,1) −8083.73 ✓, ARMA(3,0) −8083.68, ARMA(1,1) −8081.53 ✓, ARMA(2,0) −8075.06, ARMA(1,0) −8036.14. ARMA(2,1) and AR(3) are essentially tied (ΔAICC = 0.05, well below the rule-of-thumb threshold of 2). We pick **ARMA(2,1)** as the main model and use AR(3) as a Yule–Walker check.

Chosen model. ARMA(2,1) MLE: $\hat{\phi}_1 = +0.441$ (0.094), $\hat{\phi}_2 = +0.279$ (0.082), $\hat{\theta}_1 = +0.512$ (0.091), $\hat{\sigma}^2 = 1.24 \times 10^{-3}$. The constant is −0.0004 (0.004) and not significant ($p = 0.93$), which makes sense because $\hat{Y}_t$ is mean-zero by OLS construction. The AR roots are $z \in \{1.26, -2.84\}$ and the MA root is −1.95, all outside the unit disc. **Yule–Walker** (B&D §5.1.1) on the AR(3) candidate gives $\hat{\phi}^{\mathrm{YW}} = (+0.948, -0.206, +0.075)$, $\hat{\sigma}^2 = 1.25 \times 10^{-3}$, which matches the AR(3) MLE $(+0.954, -0.207, +0.071)$ to three decimals. The two estimators target the same population quantity, so they agree. **Residual diagnostics.** Ljung–Box on the ARMA(2,1) residuals with the B&D §5.3 correction df $= h - p - q$ gives $Q(20) = 43.7$, df = 17, $p = 4 \times 10^{-4}$, so the model fails the residual whiteness test. The remaining correlation sits at multiples of 7 (Figure **??**c). This means the harmonic weekly basis catches the deterministic weekday *shape* but not the shared noise between same weekdays. We report the failure rather than add more parameters at this stage.

4 Forecasting

We hold out the last 30 days (2020-09-01 to 2020-09-30) and refit ARMA(2,1) on the first $n - 30$ residuals. The h-step-ahead forecast and its prediction MSE $v_n(h)$ are computed from the ARMA recursion (B&D §3.3.1; `statsmodels.get_forecast` runs the innovations algorithm internally). The Gaussian 95% PI $\hat{Y}_{n+h} \pm 1.96\sqrt{v_n(h)}$ is then transformed back to MW by adding the extrapolated $\hat{m}_t + \hat{s}_t^{(\mathrm{year})} + \hat{s}_t^{(\mathrm{week})}$ and exponentiating. The MW interval is asymmetric (log-normal), but since the transformation is monotone the 95% coverage is preserved. Exponentiating the conditional mean gives the conditional *median* of X_{n+h} (the mean differs by a factor $\exp(v_n(h)/2) \leq 1.0025$, which is negligible).

Skill (held-out 30 days). RMSE = 187.6 MW, MAE = 145.1 MW, RMSE/std(train) = 0.097. The empirical 95% PI coverage is 30/30, which is consistent with the nominal level (note $0.95^{30} \approx 0.21$). Forecast and actual values agree to within about 250 MW on every test day (Figure **??**). After a few days the residual forecast decays towards zero, so the medium-range forecast is mostly the deterministic seasonal and trend part.

5 Reflection

Difficulties. The main problem was the leftover lag-7 correlation in $\hat{Y}_t$, which a low-order ARMA cannot fully absorb. The chosen ARMA(2,1) still works well for the forecast, but a SARIMA model with a seasonal term at lag 7 would be a natural alternative that targets this leftover structure directly. **Robustness.** Results are not very sensitive to the model choice: ARMA(2,1) and AR(3) give point forecasts that differ by at most 4 MW per day and PI widths by less than 1%. The biggest forecast errors come from holidays and weather events that the seasonal model does not see. **Alternatives.** A SARIMA term at lag 7, exogenous regressors (e.g. holiday flags, heating-degree-days), or a period of 365.25 instead of 365 would all be reasonable next steps. **Software.** Python 3 with `statsmodels` 0.14 (`ARIMA`, `yule_walker`, `acorr_ljungbox`, `plot_acf/pacf`), `numpy`, `pandas`, and `matplotlib`. The pipeline can be reproduced with `jupyter nbconvert --execute project_part_a.ipynb`.

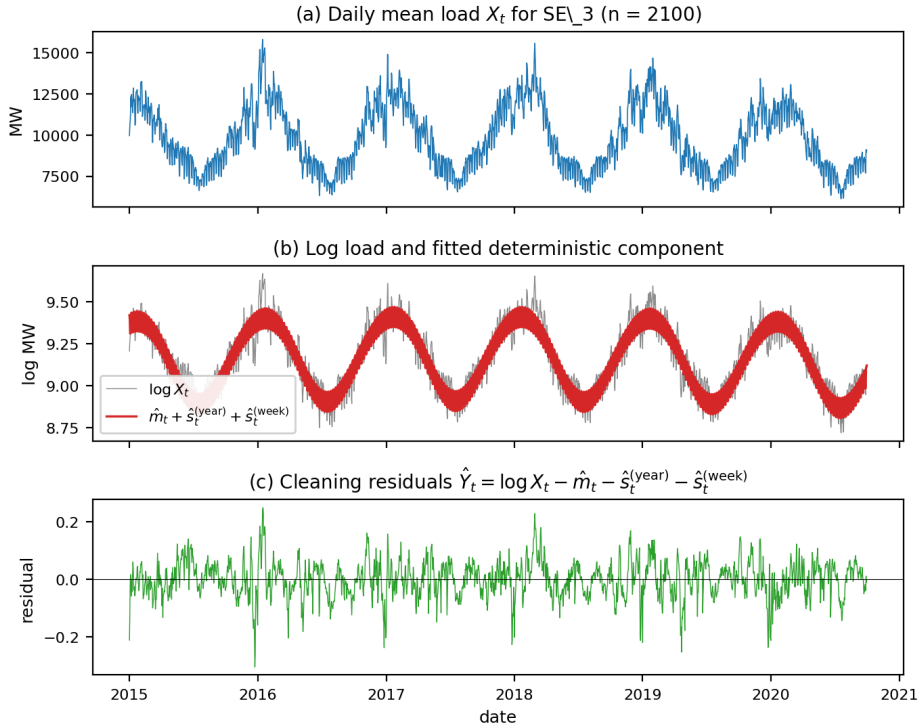

Figure 1: Classical decomposition (B&D §1.5) of daily SE_3 load, 2015-01-01 to 2020-09-30. (a) Raw daily mean load X_t, $n = 2100$. (b) Log-load $\log X_t$ (grey) overlaid with the fitted deterministic component $\hat{m}_t + \hat{s}_t^{(\text{year})} + \hat{s}_t^{(\text{week})}$ (red); seasonals are harmonic regressions ($d = 365, k = 2$ and $d = 7, k = 3$) and the trend is a quadratic OLS fit. (c) Cleaning residuals $\hat{Y}_t$ entering the ARMA stage.

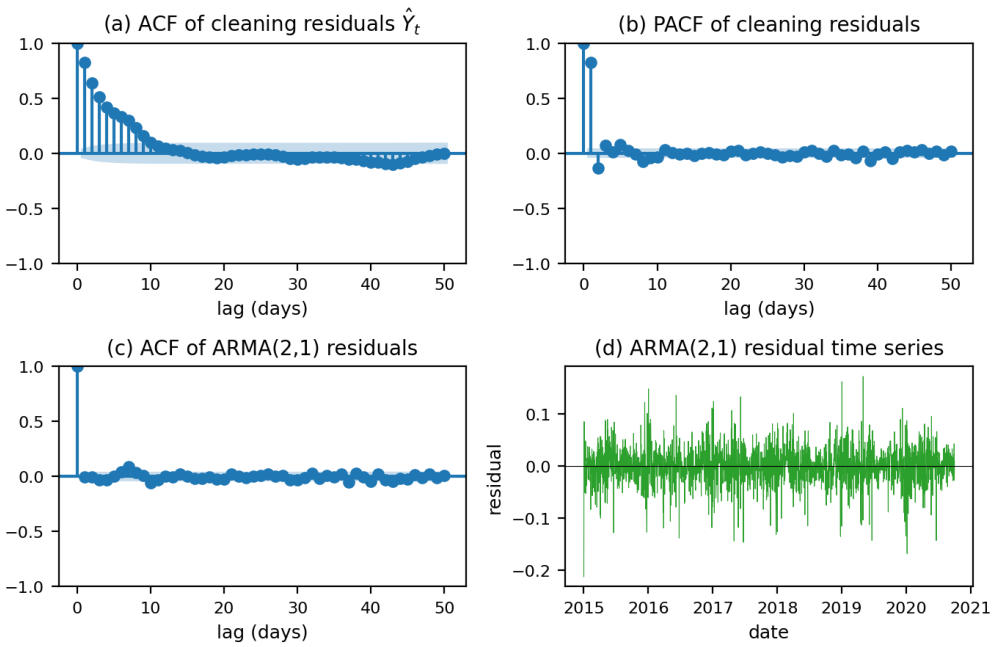

Figure 2: Diagnostic plots. (a) Sample ACF of cleaning residuals $\hat{Y}_t$ (lags 0–50, blue bands at $\pm 1.96/\sqrt{n}$, B&D §1.4.1). Slow decay $\hat{\rho}(1) = 0.83$ and the lag-7 spike $\hat{\rho}(7) = 0.30$ motivate an AR-dominated ARMA fit. (b) Sample PACF of $\hat{Y}_t$, largest spike at lag 1. (c) ACF of ARMA(2,1) residuals; the surviving correlation is concentrated at multiples of 7 ($Q(20) = 43.7$, df $= 17$, $p = 4 \times 10^{-4}$). (d) ARMA(2,1) residual time series; level and amplitude stable.

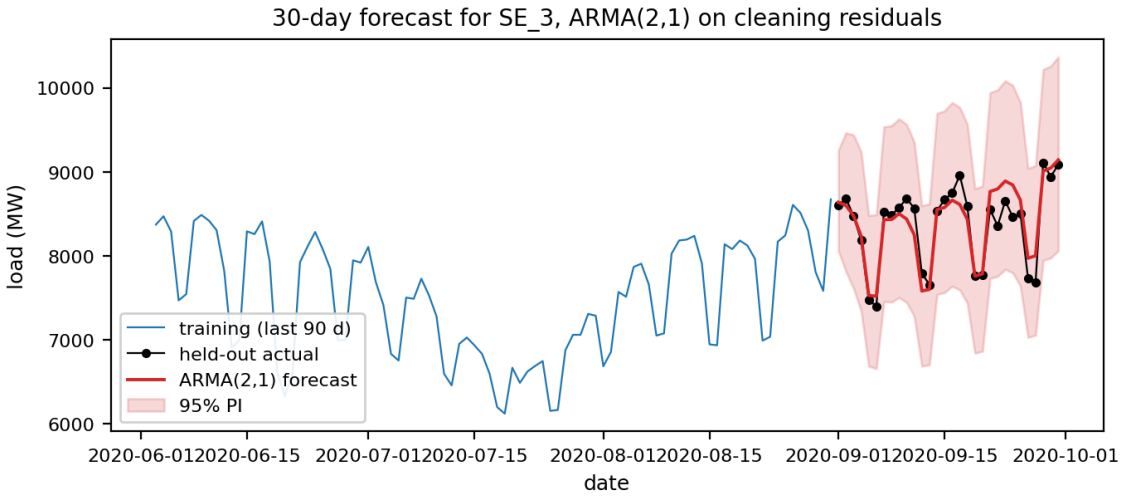

Figure 3: 30-day forecast on the held-out window 2020-09-01 to 2020-09-30, ARMA(2,1) refit on the first $n - 30$ cleaning residuals. Blue: last 90 days of training. Black markers: held-out actual daily means. Red: h-step ARMA forecast (B&D §3.3.1) inverted to MW by adding the extrapolated trend and seasonals and exponentiating; red band is the 95% PI (log-normal in MW). RMSE = 187.6 MW, MAE = 145.1 MW, empirical PI coverage 30/30.